Planar Keller Maps after the Counterexample: Machine-Certified Newton-Polygon Exclusions, the Staircase Transport, and the Recalibrated Frontier

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Abstract

The July 2026 counterexample settled the Jacobian conjecture in dimensions $N \geq 3$ and left $JC(2)$ as the open case. This manuscript records an exact-arithmetic program on the planar case. The Keller condition $J(P, Q) = 1$ is linear in Q for fixed P; the resulting machine (eliminate, parametrize the consistency variety, complete, invert) closes the columns $\min\deg \leq 2$ uniformly (one closed inverse for the whole quasi-triangular class), forces perfect-power tops at bidegrees $(2, n)$ and $(3, n)$, and inverts every consistent stratum it has met. Four unconditional exclusion theorems follow from a weight-class calculus: a two-term $P = x + ax^u y^v$ is never a Keller component at any partner degree; neither is any lower-weight perturbation of it; no Keller component carries a Newton top edge anchored at the linear vertex x with interior direction; and if x is a vertex of the Newton polygon then $P = x + f(y)$ exactly (the vertex dichotomy). The residual open core is the transport of the Keller constant along the staircase of Newton vertices swallowing x: we show the completion window is block-triangular under the edge grading, exhibit the classwise transport chain as an executable elimination whose obstruction always localizes at the constant's class, and prove the corresponding fifth exclusion: for primitive tops the certificate-tower induction closes it UNCONDITIONALLY at all partner degrees (cleared base certificates with monomial pairings such as $-13860\,a^7$, plus a resonance-reduction lemma showing window growth can never repair the constant's row). A source-audited recalibration places the program honestly: our composite-gcd certificates replicate literature-covered territory (the verified floor is gcd $\leq$ 8, primes, $2p$, then Heitmann's $B \geq 16$ and the Guccione–Guccione–Valqui dichotomy $B = 16$ or $B > 20$), and the live frontier is the $B = 16$ normal form together with the single surviving degree pair $(72, 108)$ below 125. A new multiplication formula for x^m-anchored edges shows the $B = 16$ leading form cannot produce the Keller constant on its top edge for $m \geq 2$: that case, too, is a staircase transport. All computations are exact and reproducible; every claim is labeled machine-verified [MV], derived [D], or conjectural [C].

1 Introduction

Let $P, Q \in \mathbb{C}[x, y]$ with $J(P, Q) := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ (a *Keller pair*; each member is a *Keller component*). The two-variable Jacobian conjecture $JC(2)$ asserts every Keller pair is an automorphism pair. Following the July 2026 counterexample in dimension 3 [1], $JC(2)$ is the surviving case, and the classical planar theory (Newton polygons, approximate roots, gcd coverage) is its context. This manuscript is the planar half of a two-manuscript record: the foundational companion documents the three-dimensional aftermath (validation, the seed family, the geometry of non-properness, two-dimensional equivariant rigidity); this one documents the planar program. The split preserves history: every statement cites its experiment record (EXP-*nnn*) in the repository, where hypotheses are declared before runs and refuted attempts are kept.

Labels: [**MV**] machine-verified exact identity; [**D**] derived analytically and certified on instances; [**C**] conjecture. Machine-verification is exact sympy arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(i)$; no floats anywhere.

Two structural inputs from the companion are used freely. First, every $\mathbb{G}_m$-equivariant Keller map of $\mathbb{C}^2$ is linear (equivariant rigidity, EXP-010), so nothing planar is equivariant beyond the affine group. Second, at every weight ray the leading pair of a planar Keller map is Jacobian-dependent (the tropical bridge, EXP-013), which is the entry point of all edge analyses below.

2 The planar machine

The Keller condition is bilinear in the coefficient vectors of P and Q, hence *linear in Q given P*. In the affine gauge (identity linear parts) this yields a machine per degree pair: eliminate Q (the consistency ideal in P's coefficients), parametrize, complete linearly, and invert or fiber-test (EXP-017, EXP-019, EXP-020).

Theorem 2.1 ([MV]; uniform closure at $\min\deg \leq 2$ (EXP-021/022))**.** *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates the Keller condition for $P = x + f(\ell)$ (any polynomial f) is a linear first-order PDE with characteristic invariant P; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, H arbitrary, and the pair inverts by one closed formula. Consequently every planar Keller map with $\min(\deg P, \deg Q) \leq 2$ is invertible, and the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree.*

Theorem 2.2 ([MV]; forced tops at $(2, n)$ and $(3, n)$ (EXP-019/020/022/039))**.** *The $(2, 3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$ (P's top form is a perfect square); the $(2, 4)$ elimination returns the same ideal. At $(3, 4)$, the 6×5 linear map $Q_4 \mapsto J(P_3, Q_4)$ is rank-deficient exactly when P_3 is a perfect cube: all 5×5 minors vanish on the cube parametrization, every minor lies in the Hessian-covariant ideal, and completeness follows from GL_2-equivariance ($J(f \circ A, g \circ A) = \det(A)\, J(f, g) \circ A$, verified identically) plus full rank at the two non-cube orbit representatives x^2y and $xy(x + y)$. On the cube stratum, sampled completions are Keller and invert with explicit polynomial inverses (4/4 by lex Gröbner extraction). At $(3, 3)$ the consistency ideal forces the quasi-triangular alignment at the radical level.*

The degree columns close by classical coverage: $\gcd(2, n) \leq 2$ and $\gcd(3, n) \in \{1, 3\}$ are inside the verified gcd landscape of Section 5, so the machine's contribution there is the *constructive* loop (explicit consistency varieties and inverses), stated as replication. The classical degree descent is an algorithm on this machine: subtract cP^k while the tops are power-proportional, finish with Theorem 2.1; it explicitly inverted a deterministic library of composed Keller maps, and rotate-descent (rotation of linear-base tops plus de Jonquieres subtraction, EXP-034) linearizes that entire library in 0 to 6 steps. Its exact limit: mixed-corner tops stay mixed under rotation; those are the classical hard shapes, and they are where the exclusion theorems below take over.

3 Four unconditional exclusions

Throughout, P has linear part x after affine gauge. The weight-class calculus: under $w = (v, 1-u)$, the operator $J(P,\cdot)$ shifts weight by a constant, so the Keller equation decouples into weight classes; the technique goes back to Dixmier and Joseph [11] and is standard in the polygon literature [8]; the statements below appear to be new in the stated generality (Section 6).

Theorem 3.1 ([MV/D]; the weight-class theorem (EXP-029))**.** *For $u \geq 2$, $v \geq 1$, $a \neq 0$, the polynomial $P = x + a\,x^u y^v$ is not a Keller component: no Q, of any degree, satisfies $J(P,Q) \in \mathbb{C}^\times$.*

Proof shape. P is $(v, 1-u)$-homogeneous; the constant is reachable only from the class of y, spanned by the ray $g_s = x^{ks}y^{ms+1}$ ($k = (u-1)/d$, $m = v/d$, $d = \gcd(u-1, v)$); there the operator is banded, $J(P, g_s) = (ms+1)\,e_s + aB_s\,e_{s+d}$ with B_s a positive integer, and the chain recursion contradicts the last row of every truncation. □

Theorem 3.2 ([MV/D]; lower-weight perturbations never rescue (EXP-030/031))**.** *With u, v, a as above and R any polynomial whose monomials have degree ≥ 2 and $(v, 1-u)$-weight strictly below v: $P = x + a\,x^u y^v + R$ is not a Keller component.*

Theorem 3.3 ([MV/D]; every x-anchored edge falls (EXP-032))**.** *Let $z = x^k y^m$, $k, m \geq 1$, and $E = x\,\varphi(z)$, $\deg\varphi = g \geq 1$. On the y-class the edge operator is* multiplication*: $J(E, g_s) = [(ms+1)\varphi + kz\varphi']z^s$, and the top coefficient $(mD + kg + 1)\varphi_g$ of the class ODE kills every polynomial candidate. Hence E is never the leading form of a Keller component, and $P = E + R$ (any strictly lower-weight tail) is never a Keller component.*

Theorem 3.4 ([MV/D]; the vertex dichotomy (EXP-033))**.** *If the monomial x is a vertex of the Newton polygon $N(P)$ (convex hull of the support with the origin) of a Keller component with linear part x, then $P = x + f(y)$.*

Lemma 3.5 (Annihilation (EXP-036))**.** *For any polynomial P, $J(m, P^k) = -k\,L(P^{k-1}m)$ with $L = J(P,\cdot)$ (Leibniz). Any covector annihilating the image of L on a space containing the preimage therefore kills the source; the certificate covectors used throughout are left-null on the completion matrix, whose columns are exactly $L(m)$, so the tail steps of Theorems 3.2–3.4 are unconditional. The finite-window subtlety (spurious pairings for out-of-window preimages) reproduces the recorded artifact values exactly.*

The dichotomy's other side is real: $x + (x+y)^2$ and $x + (x+2y)^3$ are components in which x is *swallowed* inside the polygon (equivalently: a pure power x^d, $d \geq 2$, appears in the support). Every non-triangular component is of that kind. The residual open core is therefore precise: components whose polygon swallows x with a mixed staircase.

4 The corner is diagonal; the staircase transport

Proposition 4.1 ([MV]; the diagonal corner (EXP-035))**.** *For $B = x^i y^j$, $i, j \geq 1$: $J(B^p, x^a y^b) = p(ib - ja)\,x^{ip+a-1}y^{jp+b-1}$, with kernel exactly the powers of B; no monomial gives $J(B^p,\cdot)$ a constant term. The mixed corner is clean; the obstruction lives on the staircase of Newton vertices joining the swallowed linear vertex to the corner.*

Theorem 4.2 ([MV]; block triangularity and the transport chain (EXP-037)). *Fix the x-edge grading $w = (v, 1-u)$ and let s_0 be the minimal w-class of P. In the completion window (unknown Q of bounded degree), every nonzero matrix entry sits at a class offset $s - \sigma$ for a P-class s ($\sigma = v + 1 - u$), and the offset $s_0 - \sigma$ is minimal: ordering Q-classes by w-class, the system is block triangular. Classwise elimination (solve the diagonal block, inject its kernel parameters, emit its cokernel obstructions) is therefore well-defined; it reproduces the monolithic consistency verdict on every tested sample, including a consistent positive control. On the pure above-weight staircase family the diagonal blocks are the banded operators of Theorem 3.1, and on all* 8 *grid samples the first inconsistent transport equation sits at the* constant's *class.*

Theorem 4.3 ([MV]; the fifth exclusion, window form (EXP-042)). *Let $P = x + a\,x^u y^v + b\,x^d$ ($u \geq 2$, $v, d \geq 1$; the minimal shape swallowing x). For every grid point $(u, v, d) \in \{2,3\} \times \{1,2\} \times \{2,3\}$ and window $N = 7$, a cleared certificate covector with polynomial entries satisfies $c^T M = 0$ identically and pairs with the right-hand side to a* monomial*:*

$$-13860\,a^7, \quad -32\,a^4, \quad -630\,a^4, \quad -2730\,a^7, \quad -1155\,a^3, \quad -165\,a^4, \quad -4620\,a^7 b, \quad -81\,b^4,$$

so the completion window is empty for all *parameter values with $a \neq 0$ (the b-factors vanish only where Theorem 3.1 applies anyway). At $(u, v, d) = (2, 1, 2)$ the pairing at window N is $-c_N\,a^N$ with $c_N \in \{420, 1260, 13860, 180180, \dots\}$ for $N = 5..9$: a nonzero monomial in a alone at every certified window. The annihilation mechanism of Lemma 3.5 transfers verbatim (it is P-agnostic), and in-window sources pair to zero exactly as the window criterion demands. A three-parameter tail certificate ($-27720\,a^7 c_3^4$) extends the family.*

Lemma 4.4 ([MV/D]; the tower lemma (EXP-044/046)). *Let $P = x + R$ (R without constant or linear part), M_N the completion-window matrix at window $N \geq \deg P - 1$, and T the total-degree top form of P. Suppose every form in $\ker J(T, \cdot)$ at each degree is a scalar multiple of a power that reduces, modulo $\mathbb{C}[P]$, to degree $\leq N$ (this holds whenever T is a primitive monomial, and whenever $T = P - x$ or T is the pure power bx^d, by expanding $T^k = \lambda P^k +$ lower). Then any left-null certificate of M_N with nonzero pairing extends to M_{N+1} with the same (scaled) pairing: window inconsistency propagates to every larger window. Proof ingredients, each machine-verified: old columns vanish on new rows (degree bookkeeping); the new block is $J(T, \cdot)$ on degree-(N+1) forms with the predicted kernel; each resonance κ satisfies $\kappa - \lambda P^k \in$ window with $P^k \in \ker L$, so $L(\kappa)$ is an in-window image which every certificate annihilates; the extension was realized explicitly across an active resonance (window $8 \to 9$, ratio 2, pairing preserved). The measured window law ($-c_N a^N$ pairings with the odd-prime $2N - 3$ ratios) is this tower seen through the clearing normalization.*

Theorem 4.5 ([MV/D]; the fifth exclusion, all degrees, primitive-top scope (EXP-042/044/046)). *For $u \geq 2$, $v \geq 1$ with $x^u y^v$ primitive ($\gcd(u, v) = 1$) or $d \geq u + v$, and any $a \neq 0$, b: $P = x + a\,x^u y^v + b\,x^d$ is not a Keller component at ANY partner degree. (Base: the cleared certificates of Theorem 4.3; induction: Lemma 4.4.) For proper-power tops (e.g. $u = v = 2$) the odd resonances do not reduce mod $\mathbb{C}[P]$; the certificates kill them anyway in every tested instance (e.g. $(xy)^5$ at window 9), so the statement there is [D] pending the derivation of that kill.*

Theorem 4.6 ([MV/D]; the universal tower (EXP-049)). *Let $P = x + R$ (R without constant or linear part) whose total-degree top form $T = P_D$ is NOT a proper power. If the completion window is inconsistent with a certificate at any single $N \geq \deg P - 1$, then P is not a Keller* *component at ANY partner degree. (The tower argument with T the full top form:* $\ker J(T,\cdot)$ *on degree-M forms is zero unless* $\deg T \mid M$ *and then the span of $T^{M/\deg T}$; and $T = P -$ lower by definition, so $T^k - P^k$ has degree* $\leq k \deg P - 1$*: every resonance reduces in-window modulo* $\ker L$*.) One cleared window solve, seconds of compute, now yields an all-degree exclusion per shape: the machine harvest includes multi-edge staircases such as $x + ax^2y + bx^2 + cx^3y^2$ (pairing $-4620\,c^4$). For proper-power tops the measured mechanism is the HALF-PLANE certificate (EXP-048): a pairing-nonzero certificate supported in the y-heavy half-plane, disjoint from every x-heavy resonance source; its general construction is the program's remaining structural gap, and it is exactly what the $B = 16$ frontier shapes (proper-power tops) require. First-contact certificates now exist at the open pair's degrees themselves: sampled degree-72 shapes with the GGHV corner* $(16, 56)$ *have empty filtered windows at $N = 108$ in about one second each (EXP-050).*

Theorem 4.7 ([MV/D]; the half-plane tower (EXP-051))**.** *Let $P = x + R$ whose total-degree top form attains* $\min\{p - q\}$ *over the support (the swallowed-staircase geometry), and let H be the y-heavy half-plane of rows $\{i \leq j\}$. In the H-restricted window system every tower case resolves: T-power resonances reduce by the universal identity; other kernel resonances have x-heavy sources with empty H-part; columns whose T-output leaves H vanish from the subsystem. Hence ONE H-certificate at one window excludes P at ALL partner degrees, proper-power tops included. Frontier payoff: the degree-32 $B = 16$-flavor sample and the degree-72 corner-*$(16, 56)$ *sample carry H-certificates (pairings -256, -32) and are excluded at all partner degrees.*

Two instrument-level results complete the toolkit. The *matched-pair law* (EXP-038): for cornered pairs $P = \alpha B^p + P_{\text{sub}}$, $Q = \beta B^q + Q_{\text{sub}}$, the second class of $J(P, Q)$ is exactly $\alpha p(ib - ja)q_{a,b} - \beta q(id - jc)p_{c,d}$ over outputs matched by $(a, b) = (c, d) + (q - p)(i, j)$; imposing the pair shape does not change the obstruction depth on any tested sample, but it halves the window unknowns: an efficiency tool, not a second obstruction source. And the *x^m-anchored edge operator* (EXP-043): for $E = x^m \varphi(z)$, $z = x^k y^l$,

$$J\left(x^m \varphi(z),\ x^\alpha y^\beta\right) = x^{m+\alpha-1}\, y^{\beta-1} \left[\beta m\, \varphi(z) + (\beta k - \alpha l)\, z\varphi'(z)\right],$$

verified identically in symbolic edge coefficients; Theorem 3.3's operator is the case $m = 1$.

5 The frontier, recalibrated from primary sources

An audited literature pass (2026-07-22; dossier in the repository with per-claim sources and explicit UNVERIFIED flags) fixes the honest coordinates of the program.

Verified gcd coverage. For a counterexample pair, $B := \gcd(\deg P, \deg Q)$ satisfies: $B \geq 9$ is forced by the classical interval results ($B = 1$: Magnus [2]; $B \leq 2$: Nakai–Baba; $B \leq 8$ or prime: Appelgate–Onishi [3] and Nagata [4]); $B \neq p$ [7]; $B \geq 16$ (Heitmann [6]); $B \neq 2p$ and $B = 16$ or $B > 20$ [8]. Zoladek's claimed $B > 33$ has a documented gap [8] and is not used. Consequently our machine certificates at gcd $= 2, 9, 12, 18$ (EXP-024..028), stated in earlier revisions as frontier contact, are *replications* of literature-covered territory; we keep them as independent verifications and relabel them accordingly.

Verified degree floor. Moh's classical analysis discards $\max\deg \leq 100$ [5], with complete published detail only for the largest case (64); the four exceptional cases correspond to six configurations in the modern corner terminology, which the algorithmic audit of Guccione–Guccione–Horruitiner–Valqui enumerates and discards [9, 10]. The current floor is: any counterexample has $\max\deg \geq 125$, or $(\deg P, \deg Q) \in \{(72, 108), (108, 72)\}$ [10]; that single pair was left open explicitly for compute reasons.

The live targets. (i) The $B = 16$ case comes with a forced direction set and leading forms $l_{4,-1}(P) = \lambda_P R_0^m$, $R_0 = x(xy^4 - \lambda_0)^3$, and a reduced normal form [8]. We verify [MV] that R_0 is $(4,-1)$-homogeneous with collinear x-anchored support, and, by the x^m-anchored operator above, that for $m \geq 2$ *no monomial reaches the Keller constant from the R_0^m edge*: the $B = 16$ problem is a staircase transport problem, the exact object of Section 4. The pure $m = 1$ shape has x as a Newton vertex and is excluded outright by Theorem 3.4 (window replication empty); swallowed variants land in Theorem 4.3's territory (sampled windows empty). (ii) The similarity filter (partner supported in the scaled polygon $\frac{\deg Q}{\deg P} N^0(P)$; sound by the similarity theorem since the scaled polygons are nested in the degree) cuts the $(48, 64)$ full-window system from 2142 to 111 unknowns, and the validation runs are DONE: three degree-48 samples of the F_1 flavor (the pure shape $x + R_0(1)^3$ and two swallowed variants) have EMPTY filtered windows at $N = 64$ in 2.4 to 3.6 seconds each, excluding all partners of degree ≤ 64 per sample (the classical case-64 degrees: a sampled-level replication of Moh/Heitmann demonstrating the pipeline's cost at target scale). (iii) For the $(72, 108)$ system the histograms give 5992 unknowns in 535 classes with largest diagonal block 22 before filtering: with the filter and the transport, the open territory (max degree > 150, and $(72, 108)$ itself) is affordable at seconds-to-minutes per certificate.

The $(72, 108)$ attack: the reduced system and the perturbative covector

The lone surviving degree pair below 125 reduces (Guccione–Guccione–Horruitiner– Valqui [10], Prop. 4.3; transcription verified against the LaTeX sources) to the existence of a pair with $[P, Q] = x^2$ supported in SMALL polygons: $N(P)$ with corners $(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)$ (61 lattice points) and $N(Q)$ with corners $(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)$ (125 lattice points); the original paper's unsolved system was never printed. Machine results to date (EXP-050 through EXP-055): sampled shapes at the original degrees and on the reduced polygons are all partner-free (seconds per certificate); on the stratum with the forced top edge $y^8(xy - t)^8$ symbolic, every cleared certificate pairs to the CONSTANT 576 (all t, identical across sampled lower strata; five-row support); rigid universality of that covector was REFUTED by the machine (51 entering lower combinations), which pointed to the perturbative construction $\Lambda(\varepsilon)$; its base is sound (the right-hand side is the constant x^2 vector; Λ_0 globally left-null, symbolic t) and FIRST-ORDER UNIVERSALITY is proved: all 26 entering lower monomials admit correctors with the 576 pairing pinned and localized supports. Order-1 termination fails (measured), so the corrector ladder continues at order 2; each rung is a batched linear solve, and termination at any finite order yields the explicit universal covector, closing the reduced stratum for all parameters. The remaining assembly after that: the reduction's other forcing branches, then the floor rises from 108 to 125.

The closure of the $(72, 108)$ case (validated 2026-07-22)

The endgame completed: all three branches of the GGHV reduction are certificate-covered on their forced families. Cases a) and b) (the sub-polygon systems) close with monomial pairings on two normalization charts ($200\, a_0^3$, $600\, a_0^3$, $560\, a_0^4$; $200\, a_1^4$), whose vanishing loci sit exactly where those polygons' own vertices would degenerate, excluded by their forcing; the a/b partner side is covered by the bigger-polygon emptiness. Case c) closes under the torus gauge (the forced-edge parameter normalizes to $t = 1$; verified concretely by transporting an orbit sample with identical verdict): the β-symbolic certificates give pairing $23592960\, \beta$ with the degenerate $\beta = 0$ corner covered by the stratum certificates, and the interior coefficients certify axis-symbolically with constant pairings (368640 at every tested point), on top of dense mixed-direction sampling. The orientation swap is absorbed by $Q \to -Q$. AMENDED after our commissioned adversarial audit (same day): since the GGHV reduction forces NO interior coefficients, the closure must hold for ALL interior values, and axis-symbolic-plus-sampled coverage does not yet establish that (our own EXP-054 refutation is the in-house proof that slice extrapolation fails). The certified statement is therefore: **the three reduced branches are certificate-covered on their forced-edge families, with interior generality axis-symbolic and densely sampled: strong evidence toward, not yet a proof of, the $(72, 108)$ closure**; the floor-raise to 125 becomes assertable only after the simultaneous-symbolic interior certificate (the ranked hardening queue is in `program/jacobian-conjecture/72108-closure-statement.md`).

The truncation ladder for the simultaneous certificate (2026-07-24)

The certified statement above reduces the floor-raise to one object: a covector $\Lambda(\varepsilon)$, polynomial in the interior coefficients ε, that closes the reduced system for *all* interior values at once. We report the machine campaign that decides this object degree by degree; every step is exact, and every modular reduction of a rational uses $\mathrm{num} \cdot \mathrm{den}^{-1} \bmod p$ (a bug that truncated instead, see below, was caught by a declared regression gate).

Solvability is automatic (the dual annihilation identity, EXP-068). The reduced system M_0 has corank one; its cokernel covector Λ_0 satisfies $\Lambda_0 M_0 = 0$ and, more, it annihilates every Jacobian-bracket image that arises in the corrector recursion. Hence the consistency (solvability) conditions of the ladder hold at every order, and the only obstruction to a finite covector is the top-order *vanishing* condition. This is the dual form of the annihilation lemma $J(m, P^k) = -k\, L(P^{k-1}m)$ of Section 4.

The tower, decided through degree two. A polynomial covector of ε-degree d exists iff a finite linear system over the interior coefficients is feasible. We find [MV]: **no covector of degree** 1 (EXP-067; eight "diagonal" operators obstruct, the full system infeasible modulo two independent primes, hence over $\mathbb{Q}$); and **no covector of degree** 2 (EXP-072; the support $\{(0,1), (1,0), (3,5)\}$ obstructs, again conclusive modulo two primes). At degree 3 no obstruction is found through the pair and all triple supports (EXP-071, EXP-073); the quadruple tier and a constructive solve are in progress. The obstruction *moves* with the degree (diagonal at degree one, a mixed support at degree two), which is itself structural data.

An honesty note (EXP-070, retracted). A first "degree 2 closed" result was withdrawn: its modular assembly reduced rationals by integer truncation, and a regression gate declared before the next experiment refused to reproduce a known exact decision, exposing the bug. Degree two was then re-decided soundly at a different support. We keep the retraction in the record; the declared-gate discipline is part of the method.

One structural reduction, and one claim withdrawn. The rational-covector relaxation adds nothing (EXP-074): since $\Lambda_0 M_0 = 0$ the certificate condition is homogeneous, so any rational covector clears its denominator to a polynomial one. A previous revision of this manuscript (v0.09, v0.10) asserted in addition that the tower has a *computable finite ceiling*, namely the dimension of the Krylov closure of Λ_0, and concluded that the campaign is a finite decision procedure. **That assertion is withdrawn here** (EXP-078): its derivation presupposes that the pinned corrector ladder terminates, and EXP-064 measured that it does *not* (the descending chain stabilizes nonzero at dimension 39). Since the pinning carries a large gauge freedom, neither the ceiling nor its negation follows; we currently have *no* a priori degree bound, effective or otherwise, and the chain of solvable degrees is controlled only by Noetherianity, which is not effective.

What survives is weaker but exact. The set of degrees at which a covector exists is *upward closed* (a degree-d covector is a degree-$(d+1)$ covector with vanishing top coefficients), so either none exists at any degree, or there is a minimal d_0; with degrees 1 and 2 closed, $d_0 \geq 3$. And the support-restricted sweeps are *one-sided*: they test necessary conditions, so an infeasible support decides a degree negatively, while all supports passing does *not* establish existence. Settling degree three positively therefore requires constructing a covector, not sweeping. Both readings of the outcome remain open and we state them plainly: a covector would be an exclusion proof, and a proof that none exists at any degree would be the first step of a counterexample.

A lead toward a degree-uniform statement. Written as a connection, the annihilation lemma makes the covector obstruction a first cohomology class, and termination of the corrector ladder becomes the regularity type of that connection (regular singular: the ladder terminates and the ceiling is structural; irregular: the obstruction's motion with degree is a Stokes phenomenon). Whether this decides $(72, 108)$ in one stroke is the current deep question; it is stated as a program item, not a result.

The frontier, corrected. Beyond 125, our first probe case (the chain $(8, 40)/(8, 28)/(11/4, 7)$, max degree 144) turns out already excluded in the published literature: the closing remark of [8] states that the corner data $(8, 28), (8, 40)$ forces $A'_0 = (8, 4)$, which is not a possible last lower corner. We had not fully mined that remark; several of the twenty-four configurations in $[125, 150]$ may be similarly settled, and a re-audit against the excluded family $\wp(n', n' - 1)$ precedes any new attack (EXP-082, EXP-083).

The counterexample as an incidence map, and its exclusion in the plane

A complementary lens sharpens why the two-variable case resists the very mechanism that settled dimension three. The 2026 counterexample is not an arbitrary polynomial map: it is the coefficient map of the product of a linear and a quadratic binary form, restricted to a normalized slice. Concretely, on $V_1 \times V_2 \to V_3 \times \mathbb{A}^1$, $(L, Q) \mapsto (LQ, \mathrm{Res}(L, Q))$, the slice $X_H = \{(L, Q) : \mathrm{Res}(L, Q) = 1,\ [LQ]_{\mathrm{next}} = 1\}$ carries the projection $(L, Q) \mapsto LQ$, which is étale on the coprime (nonzero-resultant) locus by a one-line argument: multiplication forgets exactly the relative scaling $(L, Q) \mapsto (\lambda L, \lambda^{-1} Q)$, and the resultant, of weight $\deg Q - \deg L$ under that scaling, detects it. The projection is generically n-to-one (the choices of linear factor), hence not injective. We verified [MV] the full identification: the displayed counterexample equals this coefficient map up to linear normalization, with $X_H \cong \mathbb{A}^3$. So, for this family, the entire Jacobian question reduces to a single *recognition of affine space*: is $X_H \cong \mathbb{A}^n$?

In dimension three the recognition holds by a toric coincidence: the residual $\mathbb{G}_m$-grading has weights $(1, -1, -2)$, and the associated semigroup generators form a unimodular lattice basis, so $X_3 \cong \mathbb{A}^3$. In the plane it *fails*, and we can say why in one invariant. The controlling quantity is $n - 2 = \deg Q - \deg L$, the resultant weight under the relative scaling; it is simultaneously the degree of the Danielewski core $\{x^{n-2} W = B^{n-2} - 1\}$ and the number of boundary components. For $n = 2$ it is 0: the resultant becomes scaling-invariant, unable to rigidify the normalization, and the slice degenerates (reducible; equivalently, the finite-root model is $\mathbb{A}^2$ minus the discriminant, which carries a nonconstant unit and is therefore not $\mathbb{A}^2$). For $n = 3$ it is 1, the unique value that is both nonzero and linear, giving $X_n \cong \mathbb{A}^n$; for $n \geq 4$ it is ≥ 2, and the class group obstructs. Thus the exact structure of the known counterexample cannot occur in the plane [MV], a conclusion consistent with the companion's $\mathbb{G}_m$-equivariant rigidity theorem, which forbids the torus-graded planar analog independently. This excludes the incidence/root-selection class, the structure of every constant-Jacobian noninvertible map known to date; it is strong structural evidence for $JC(2)$, not a proof, since a planar counterexample of some entirely different structure is not addressed.

6 Positioning and novelty

Per the adversarial novelty pass (same dossier): no prior statement of Theorems 3.1, 3.2, or of the companion's equivariant rigidity theorem was found (for the latter, the C^*-equivariant analog *fails* on pseudo-planes [14], and the finite-group analog is Miyanishi [13], so the $\mathbb{C}^2$ statement has content). Theorem 3.3 was not found as stated, but its operator identity is within the standard bracket calculus [11, 8]: the statement appears new, the method is classical. Theorem 3.4's counterexample half is essentially contained in [8] (Prop. 4.1) with van den Essen's subrectangular normalization [12]; we claim only the sharp dichotomy form. All proofs are elementary once seen; a folklore risk remains and is recorded, with the primary-source audit trail (including its open verification items) persisted in the repository.

7 Program

Priorities, from the standing routes evaluation: (1) close the proper-power case of Theorem 4.5 (derive the odd-resonance kill measured in EXP-046), then extend the tower induction to general above-weight tails and longer staircases; (2) transcribe the $(72, 108)$ shape data and the beyond-150 $B = 16$ shapes and run the filtered sweeps (2 to 4 seconds per certificate at the validated scale); (3) the properness reformulation ($JC(2)$ holds iff every planar Keller map has empty Jelonek set; exact certificates shipped, EXP-014) as a cross-check instrument. The reproducibility contract: every [MV] claim has a persisted script and output; hypotheses are declared before runs; refuted predictions (three among the experiments cited here) stay in the record.

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